Supplementary material for "Evaluating Authorization Regimes and Runtime Guardrail Configurations for Tool-Using LLM Agents"

These twelve tables provide detailed results referenced by the main manuscript. They introduce no additional hypothesis family or claim beyond the analyses reported in the article.

Table S1: Online-state source audit

Conclusion	Audited units	Independently reviewed	Claim use
No online gold observed on covered path	9	0	Path-scoped only
Not a runtime guard	10	0	Taxonomy only
Mixed or path dependent	1	0	Path-scoped caution
Inconclusive	2	0	No substantive claim

Table S2: Completed medium security and utility outcomes

Condition	Unsafe effects / 36 attacks	Unsafe-effect rate (Wilson 95% CI)	Tool completions / 24	Tool TCR (Wilson 95% CI)	No-tool TCR
No defense	8/36	22.22% (11.72–38.08)	12/24	50.00% (31.43–68.57)	12/12
Safe prompt / spotlighting	8/36	22.22% (11.72–38.08)	11/24	45.83% (27.89–64.93)	12/12
PromptGuard (detector-only)	8/36	22.22% (11.72–38.08)	12/24	50.00% (31.43–68.57)	12/12
Structural baseline	8/36	22.22% (11.72–38.08)	12/24	50.00% (31.43–68.57)	12/12
Policy package (`m3_policy_only`)	0/36	0.00% (0.00–9.64)	7/24	29.17% (14.91–49.17)	12/12
Full package (`m3_full`)	0/36	0.00% (0.00–9.64)	6/24	25.00% (12.00–44.90)	12/12

Table S3: Prespecified paired comparisons

Comparison	Treatment minus baseline	Exact McNemar p	Holm-adjusted p	10,000-cluster bootstrap 95% CI
Full package vs structural, unsafe effect	−22.22 pp	0.0078125	0.03125	−36.11 to −8.33 pp
Full vs policy package, unsafe effect	0.00 pp	1.0	1.0	0.00 to 0.00 pp
Full package vs structural, tool TCR	−25.00 pp	0.03125	0.09375	−41.67 to −8.33 pp
Full vs policy package, tool TCR	−4.17 pp	1.0	1.0	−12.50 to 0.00 pp

Table S4: Runtime burden and operational quality

Condition	ASK / 72	Benign ASK rate	Operational failures / 72	Provider calls	Trace-bundle cost (CNY)	Total latency p95
No defense	0	0.00%	4	181	0.1821176	16.924 s
Safe prompt / spotlighting	0	0.00%	5	167	0.1812942	15.971 s
PromptGuard (detector-only)	0	0.00%	5	180	0.0986064	18.249 s
Structural baseline	0	0.00%	4	249	0.2568822	21.251 s
Policy package	16	19.44%	1	112	0.0548220	7.897 s
Full package	22	22.22%	2	117	0.1441772	10.701 s

Table S5: Public-primary benchmark strata

Benchmark / endpoint	Structural baseline	Policy package	Full package
AgentDojo attacks, unsafe effect	87/300 (29.00%)	13/300 (4.33%)	29/300 (9.67%)
M3-v2 attacks, unsafe effect	46/300 (15.33%)	0/300 (0.00%)	9/300 (3.00%)
AgentDojo benign tool, TCR	67/97 (69.07%)	13/97 (13.40%)	13/97 (13.40%)
M3-v2 benign tool, TCR	36/71 (50.70%)	36/71 (50.70%)	36/71 (50.70%)
M3-v2 benign no-tool, TCR	79/79 (100%)	79/79 (100%)	79/79 (100%)

Table S6: Defense-aware adaptive attack

Model/panel	Budget	Tasks	Unsafe-effect successes	First-success distribution	Saturated?
Primary exploratory cost canary	50	8	0 (0.00%, 95% CI 0.00–32.44%)	No successes	No
Primary frozen panel	50	40	4 (10.00%, 95% CI 3.96–23.05%)	4 first successes by query 50	No
Primary frozen panel	200	40	18 (45.00%, 95% CI 30.71–60.17%)	14 additional first successes at queries 51–200	No

Table S7: E1 runtime burden and cost

Cell	ASK incidence	DENY incidence	Operational failure	Latency p95	Tokens	Cost (CNY)
Policy / normal	65/180	15/180	15/180	8.11 s	305,672	0.9651852
Policy / safe	55/180	16/180	16/180	9.12 s	334,370	0.8439636
Full / normal	63/180	22/180	16/180	8.26 s	306,306	0.7615200
Full / safe	62/180	21/180	16/180	9.13 s	327,845	1.0896174

Table S8: E2 source-authority decisions

Evidence layer / cell	ALLOW	ASK	DENY	Operational failure	Executed actions
End-to-end policy / trusted	—	60/60 records	0/60	0/60	0/60
End-to-end policy / untrusted	—	53/60 records	0/60	0/60	0/60
End-to-end full / trusted	—	57/60 records	0/60	0/60	0/60
End-to-end full / untrusted	—	0/60	55/60 records	1/60	0/60
Replay policy / trusted	60	60	0	—	—
Replay policy / untrusted	60	60	0	—	—
Replay full / trusted	60	60	0	—	—
Replay full / untrusted	30	30	60	—	—

Table S9: E3/E4 operational summary

Study	Planned/latest	Completed	Parse failure	Provider failure	Max steps	Cost (CNY)
E3 outcome-enriched Pro	240/240	185	17	38	0	1.0690866
E4 Flash replication	360/360	347	8	4	1	0.5410834

Table S10: Conditional adaptive-tail execution

Quantity	Result
Prespecified B200 survivors completed	12/12
Latest query work keys	3,600
Physical rows / superseded attempts	3,605 / 5
First unsafe effect at queries 201–500	0/12
Conditional probability, Wilson 95% CI	0.00% (0.00%–24.25%)
Operational status	3,577 completed; 21 parse failures; 2 provider failures
Total tokens	4,511,768
Attributed cost	CNY 24.5551236
Automatic retry / provider tools / real external effects	0 / 0 / 0

Table S11: E6 DeepSeek activity and operational outcomes

Condition	Unsafe effect	Operational failure	Unsafe or failure	ASK events	DENY events
No defense	17/60	12/60	29/60	0	4
Source-pinned	16/60	8/60	24/60	0	8
VIGIL					
Policy / normal	2/60	7/60	9/60	31	7
Policy / safe	1/60	2/60	3/60	21	2
Full / normal	0/60	7/60	7/60	25	18
Full / safe	0/60	3/60	3/60	19	6

Table S12: E6 paired effects and Gemini inference coverage

Evidence	Comparison or coverage	Estimate	Newcombe / cluster-bootstrap 95% CI	Exact McNemar / Holm p	Interpretation
DeepSeek	VIGIL minus no defense	−1.67 pp	Newcombe −10.48 to +7.12 pp; bootstrap −10.00 to +6.67 pp	1.000000 / 1.000000	Interval crosses zero; benchmark directions oppose
DeepSeek	Policy / normal minus no defense	−25.00 pp	Newcombe −36.95 to −14.30 pp; bootstrap −36.67 to −15.00 pp	0.000061 / 0.000122	Risk-enriched single-block contrast
DeepSeek	Full / safe minus no defense	−28.33 pp	Newcombe −40.77 to −16.81 pp; bootstrap −40.00 to −16.67 pp	0.000015 / 0.000061	Risk-enriched single-block contrast
DeepSeek	Full / safe minus VIGIL	−26.67 pp	Newcombe −39.01 to −15.39 pp; bootstrap −38.33 to −15.00 pp	0.000031 / 0.000092	Risk-enriched single-block contrast
Gemini	Scientific terminal / planned	6/180	—	—	Five response-bearing; one response-less failure
Gemini	Minimum response-complete paired coverage	0/30	—	—	Effects not estimable